research-ideas-2

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1 Project 1

1.1 Project Title:

Using BERT to Classify Firms into Emerging Risk Categories Based on 10-K Disclosures

1.2 Goal

Build a BERT-based NLP model that reads the *Item 1A. Risk Factors* section of each firm's 10-K and classifies the firm into **multiple risk categories**:

- **Climate risk** (physical / transition)
- **Cybersecurity risk**
- **Geopolitical risk**
- (Optionally) **Other risks**: e.g., supply chain, regulatory, pandemic

This classification can then be used to:

- create **binary or probabilistic firm-level labels** for risk exposure
- build **risk co-occurrence matrices**
- analyze correlations and market impact of **multi-risk profiles**

1.3 Why BERT?

Traditional approaches (e.g., keyword matching, TF-IDF) miss **semantic nuance**, **negation**, and **context**. BERT, as a transformer-based language model, can:

- understand **how** risks are described (not just mentioned)
- handle **long-range dependencies** in legal-style 10-K text
- generalize to **unseen formulations** of emerging risks

1.4 Methodology Outline

1.4.1 1. Define Emerging Risk Classes

Create a risk taxonomy (multi-label):

```
["Climate_Physical", "Climate_Transition", "Cybersecurity", "Geopolitical", "Other"]
```

Each firm/document can have multiple labels (e.g., climate + cyber).

1.4.2 2. Data Preparation

- Extract *Item 1A. Risk Factors* from 10-Ks (e.g., using the `edgar` package)
- Preprocess:
  - Chunk if too long (>512 tokens)
  - Remove boilerplate headers
- Annotate a **training set**:
  - Manually label ~500–1000 10-Ks with the risk types or use some soft labelling techniques
  - Use existing studies (e.g., Florackis for cyber, Caldara for geo) to derive heuristics

1.4.3 3. Model Training: Multi-Label BERT Classifier

Use a pretrained financial BERT (e.g., `FinBERT`, `Legal-BERT`, or `bert-base-uncased` fine-tuned on 10-Ks):

```
from transformers import BertTokenizer, BertForSequenceClassification
from sklearn.metrics import f1_score

model = BertForSequenceClassification.from_pretrained(
    'bert-base-uncased',
    num_labels=5,
    problem_type="multi_label_classification"
)
```

Use a **Sigmoid + Binary Cross Entropy Loss** for multi-label setup.

1.4.4 4. Evaluation and Output

- **Metrics**: precision, recall, F1 (macro and per-label)
- **Explainability**: Use tools like LIME/SHAP to show why a firm was classified under "cyber"
- **Thresholding**: Convert BERT output probabilities into labels (e.g., >0.5)

1.5 Use of Output in Correlation Analysis

Once classification is complete:

1. **Assign each firm a yearly binary (or probabilistic) vector** like:

   ```
   firm_risks = {"AAPL": [1, 0, 1, 0, 0], "MSFT": [1, 1, 1, 0, 1]}
   ```

2. Compute:
   - **Risk co-occurrence matrix** across all firms
   - **Time series of risk frequencies**
   - **Overlap network**: which risks co-exist at firm level
3. Use in:
   - **Portfolio-level backtesting**: how do firms with climate+cyber exposure behave in crisis periods?
   - **Copula modeling**: model joint distribution of binary vectors (e.g., multivariate Bernoulli + copula)
   - **Systemic exposure mapping**

1.6 Advanced Extensions

1.6.1 1. Fine-tuning with Weak Supervision

Use labeling functions (via Snorkel) based on known dictionaries and events to generate pseudo-labels for large-scale training.

1.6.2 2. Contrastive Learning

Train BERT to **differentiate paragraphs** about different risk types using triplet loss — more robust to overlapping risk descriptions.

1.6.3 3. Zero-shot classification

Use `facebook/bart-large-mnli` or `cross-encoder/nli-roberta-base` to classify 10-K paragraphs into risk types **without explicit labels** — allows generalization to **new/emerging risks** (e.g., AI risk, water scarcity).

1.7 Example Applications

- Identify firms that transitioned from single-risk to multi-risk exposure (e.g., from cyber only to cyber+climate)
- Study if **market prices react more strongly** to firms with multiple BERT-classified risk types
- Detect **temporal waves**: are there macro periods where climate + geopolitical risk spike together?

1.8 Summary Table

Component	Task
Model	Multi-label BERT classifier (`FinBERT`, `Legal-BERT`)
Input	10-K Item 1A text
Labels	[Climate_Physical, Climate_Transition, Cyber, Geo, Other]
Output	Binary or probabilistic exposure vectors
Uses	Correlation, portfolio analysis, copulas, systemic risk mapping

1.9 Soft Labelling with Snorkel

Instead of **manually labeling** every 10-K document with its emerging risk types (climate, cyber, geopolitical…), we can **write a set of simple rules or heuristics** — called **labeling functions (LFs)** — that assign **noisy but useful labels** to our data.

Then, we can use a tool like **Snorkel** to **combine** these noisy signals into **probabilistic labels** that can be used to **train BERT** as if we had clean ground truth.

1.9.1 Basic structure

Define **simple labeling functions**, like:

```
def lf_contains_climate_keywords(text):
    keywords = ["greenhouse gas", "emissions", "carbon", "TCFD", "net zero"]
    return 1 if any(k in text.lower() for k in keywords) else 0
```

Another LF might look for geopolitical terms, or match against known cyberattacks:

```
def lf_mentions_sanctions(text):
    return 1 if "sanctions" in text.lower() or "foreign policy" in text.lower() else 0
```

```python
def lf_mentions_data_breach(text):
    return 1 if "data breach" in text.lower() or "unauthorized access" in text.lower() else 0
```

Each function returns a **label for one class**, or `-1` if it abstains.

1.9.2 What Snorkel does

Snorkel:

1. **Runs all LFs** across your unlabeled data (e.g., 10-K paragraphs or documents)
2. **Builds a probabilistic model** of which LFs are more accurate, correlated, or noisy
3. **Generates soft labels** (e.g., 90% chance this 10-K relates to "cyber risk")
4. These soft labels can be used to **fine-tune BERT** (or any other classifier)

1.9.3 Example in context

Imagine our corpus has 1000 10-Ks. We write ~10–20 labeling functions that detect:

- climate risk keywords or phrases
- cyber risk terminology (e.g., "ransomware", "SOC", "penetration test")
- geopolitical terms (e.g., "tariffs", "military escalation", "sanctions")
- references to disasters (matched to dates from EM-DAT or NOAA)

Then:

- Snorkel generates **noisy but scalable labels** from these LFs
- We now have **10K weakly labeled documents** for **multi-label fine-tuning of BERT**

1.9.4 Advantages

- **Avoids huge manual annotation** effort
- Leverages **expert domain knowledge** via interpretable rules
- Enables to bootstrap a good BERT classifier from scratch
- Supports **multi-label setup** naturally
- Can be updated or refined as we discover new signals

1.9.5 Related insight

Snorkel is most useful when:

- We have lots of text (e.g., many 10-Ks)
- Manual labeling is slow
- We can define keyword dictionaries or weak rules
- We plan to use a **strong model** like BERT that benefits from more data, even if noisy

1.10 Use of pretrained NLI models

The other option is using **zero-shot classification** via pretrained **natural language inference (NLI)** models like `facebook/bart-large-mnli` or `cross-encoder/nli-roberta-base`. These models allow us to **classify 10-K paragraphs into predefined or emerging risk categories without any labeled training data**.

1.10.1 Goal

Use **zero-shot classification** to detect the presence of **specific emerging risks** (e.g., *climate*, *cybersecurity*, *geopolitical*, *AI risk*, *water scarcity*) in **10-K paragraphs**, without needing to train a new model or prepare labeled data.

1.10.2 What is Zero-Shot Classification?

Zero-shot classification treats classification as a **natural language inference (NLI)** problem:

> "Given a paragraph (the premise), and a label like *This paragraph is about cybersecurity risk* (the hypothesis), how likely is the hypothesis entailed by the paragraph?"

NLI models output:

- **Entailment** (yes, the label fits)
- **Contradiction** (no, it contradicts)
- **Neutral** (uncertain)

This makes the method **flexible and adaptive**, especially to **new risk types not present in training data**.

1.10.3 How It Works

1. Define the risk types as labels For example:

```
risk_labels = [
    "climate risk",
    "cybersecurity risk",
    "geopolitical risk",
    "AI-related risk",
    "supply chain risk",
    "pandemic risk",
    "water scarcity risk"
]
```

2. Load a pretrained zero-shot model

```
from transformers import pipeline

classifier = pipeline("zero-shot-classification", model="facebook/bart-large-mnli")
```

3. Classify a paragraph from a 10-K

```
text = """Our operations rely heavily on global data infrastructure and may be disrupted by unauthorize
result = classifier(text, candidate_labels=risk_labels, multi_label=True)
```

Output:

```
{
  'labels': ['cybersecurity risk', 'AI-related risk', 'geopolitical risk', ...],
  'scores': [0.92, 0.35, 0.10, ...]
}
```

The model says: "This paragraph is *very likely* about cybersecurity risk."

4. Apply across a corpus

- Chunk each 10-K into paragraphs (or sections)
- Apply the model to each paragraph
- Aggregate predictions into document-level risk vectors (e.g., max or average probability per label)

1.10.4 Use cases

- **Classify emerging risks** not covered in your training data (e.g., *AI misuse*, *water stress*)
- **Rapidly tag large corpora** (no manual labeling or training)
- Track **temporal evolution** of risk mentions (e.g., *pandemic* spikes post-2020)
- Use as a **first-stage filter** to build datasets for supervised learning

1.11 Relationship with Snorkel / Weak Supervision

Feature	Zero-shot (NLI)	Snorkel
Requires manual labels?	No	Needs labeling functions
Requires training?	Pretrained	Needs LF training and label model
Good for new risks?	Excellent	Harder — needs LF update
Scalable?	Slower (inference-heavy)	Efficient once LFs are written
Output type	Probabilistic entailment scores	Probabilistic pseudo-labels
Interpretability	High (natural language)	High (rule-based LFs)

Complementarity:

- **Zero-shot models** are great for **bootstrapping new risk taxonomies**.
- **Snorkel** is better when:
  - We want to **scale to millions of records**
  - We can codify domain rules
  - We want **structured control over labeling**

1.11.1 Hybrid Strategy (Best of Both Worlds)

1. **Use zero-shot classification to label a set of documents**
   - Treat the top-ranked risks as ground-truth
   - Get probabilistic scores
2. **Use these soft labels to train labeling functions in Snorkel**
   - For example, extract high-scoring cyber paragraphs to build term lists for LFs
   - Snorkel can then expand to a larger unlabeled corpus
3. **Fine-tune BERT on this automatically generated data**

This allows you to:

- Rapidly prototype
- Incorporate new risks
- Scale reliably with minimal manual effort

1.12 Final Notes

- Zero-shot models like `bart-large-mnli` and `nli-roberta-base` **bypass the need for fine-tuning**
- You can **easily test new hypotheses** (e.g., *Does the firm report any risk related to space weather?*)
- They're ideal for **prototyping risk taxonomies**, before committing to full-scale model training